Multiple comparisons

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Count the family. Divide alpha. Report every decision.

Apply Bonferroni with complete family sizes, inclusive cutoffs and honest missing-outcome limits.

Keep one habit: Declared family → five-test ledger → native cutoff comparison → missing-result and validity probes → exact audit → independent four-test case.

MAIN LESSON · SECTION 1

Count the complete comparison family before applying a decision rule

> Keep familywise error control distinct from effect magnitude and null truth.

Learning goals

I can explain the **mechanism** and terminology of **multiple comparisons**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve p-values, uncertainty and independent units

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Statistical tests and assumptions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Uncertainty intervals. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Experimental units and independence. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Starting explanation

- Recall Statistical tests and assumptions. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain **multiple comparisons** using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of **multiple comparisons**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Ask how many chances the analysis had to make a false rejection

> The comparison family belongs to the research question, not the attractive results.

Begin with repeated opportunities

Imagine checking five possible improvements and reporting only the most attractive result. Even when individual tests are valid, repeated opportunities can increase the chance of a false rejection somewhere in the family. Bonferroni gives each of the declared tests a smaller allowance.

It bounds the chance of at least one false rejection by the family alpha; it does not promise that five percent of this particular family will be wrong.

Original multiple-comparison explanation

Testing many hypotheses creates more chances for false discoveries. A family rule must be specified before selecting attractive p-values.

Full family and decision contract

Declared item	Complete exercise contract
Family	All five stated tests; do not discard unfavorable results
Supplied p-values	0.009, 0.012, 0.3, 0.4, 0.8; stipulated valid tests, not fitted from invented data
Family alpha	0.05, fixed before inspecting results
Per-test rule	Reject when $p \leq 0.05 / m$, including equality
What is controlled	Probability of at least one false rejection among true nulls is at most 0.05
Dependence	Bonferroni union bound does not require independent tests; individual p-values must still be valid
Not established	Effect magnitude, practical value, null truth and actual study quality

Given and missing facts

List the given facts and one fact that is not supplied.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Divide the allowance and trace every decision

> The second-smallest result changes decision under family control.

Declared Bonferroni procedure

For this exercise use Bonferroni: family alpha .05 divided by number of tests m; reject each test when its valid p-value is at most .05/m. This controls the **family error** by a bound without requiring independent tests.

All five decisions

Main test	Supplied p	Bonferroni cutoff	Decision
H1	0.009	0.010	Reject
H2	0.012	0.010	Do not reject
H3	0.300	0.010	Do not reject
H4	0.400	0.010	Do not reject
H5	0.800	0.010	Do not reject
Unadjusted comparison	H1 and H2 are ≤ 0.05	This is a different decision rule	Does not provide the same familywise bound

Understand the union-bound argument

Why the bound works: with m tests, each true null has false-rejection probability at most alpha/m when its p-value is valid. The probability of any false rejection is at most the sum of those probabilities. There are at most m true nulls, so the sum is at most m × alpha/m =

alpha. This is an upper bound, not a measured error frequency, and it does not require independence across tests.

Record assumptions and explain the visual

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Multiple comparisons — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Five valid tests form one declared family with alpha 0.05. Their p-values are .009, .012, .3, .4, .8. Use Bonferroni: compare each p-value with 0.05/5, including equality.	For this exercise use Bonferroni: family alpha .05 divided by number of tests m; reject each test when its valid p-value is at most .05/m. This controls the family error by a bound without requiring independent tests.	Threshold .01; only .009 rejected. An unadjusted .05 rule would also reject .012 and change family control.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Predict a ten-test family with four unsupplied results

> Compute the new cutoff while leaving missing test outcomes unknown.

Main task and response

- Five valid tests form one declared family with alpha 0.05. Their p-values are .009, .012, .3, .4, .8. Use Bonferroni: compare each p-value with 0.05/5, including equality. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Predict the changed picture family

- Change: The declared family has ten tests instead of six; keep the listed values unchanged.

▼ **Check after predicting**

Prediction: The cutoff becomes 0.005, and only 0.004 among the listed values still rejects.

Reason: The error allowance is divided among ten tests; all ten results must be reported, including the four not supplied here.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read family size, cutoff and complete reporting together

> Count missing outcomes without inventing their p-values.

Main five-test result

Five valid tests form one declared family with alpha 0.05. Their p-values are .009, .012, .3, .4, .8. Use Bonferroni: compare each p-value with 0.05/5, including equality.

Threshold .01; only .009 rejected. An unadjusted .05 rule would also reject .012 and change family control.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate six-test inputs and ten-test change

Six declared prompt comparisons. A team performs six valid hypothesis tests and applies a predeclared Bonferroni familywise threshold.

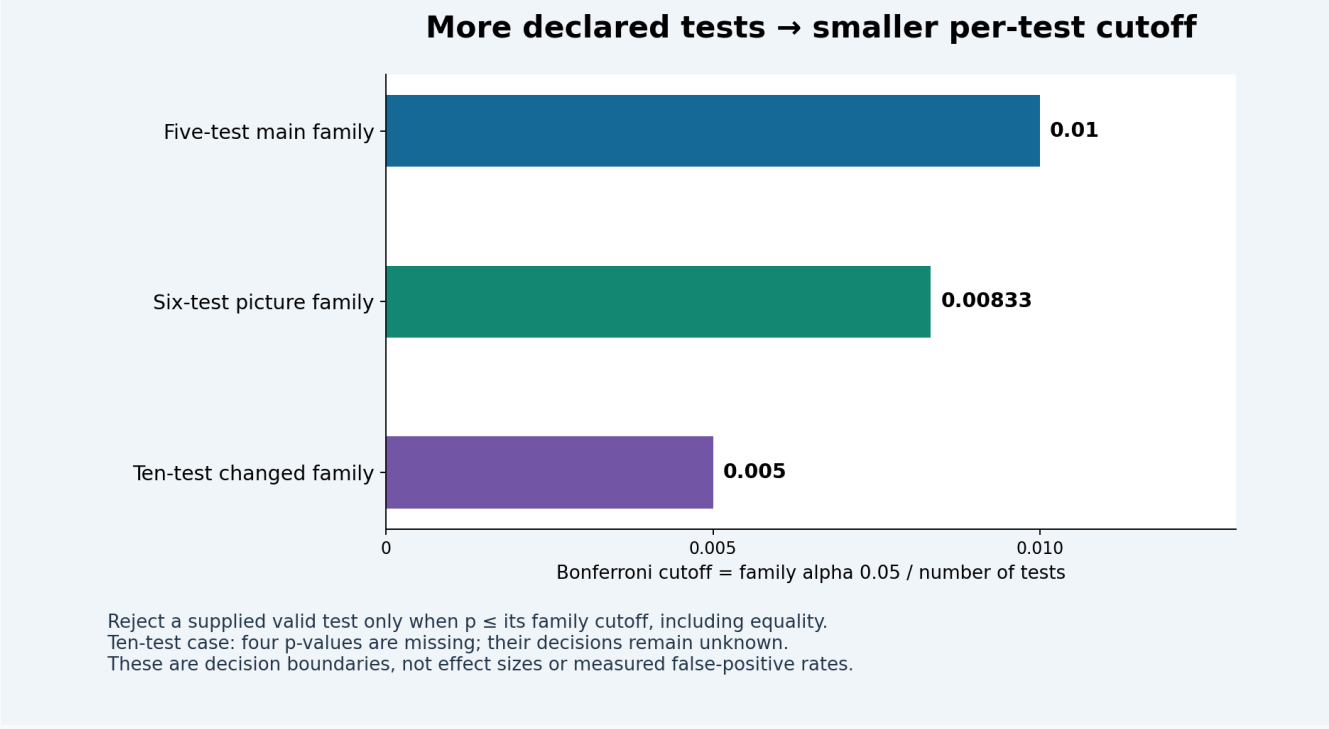
Familywise alpha=0.05. Supplied unadjusted p-values are 0.004,0.012,0.02,0.08,0.4,0.9. These values are stipulated, not calculated from invented data.

The separate picture has six supplied p-values: 0.004, 0.012, 0.02, 0.08, 0.4 and 0.9. Its exact cutoff is 0.05/6 = 1/120, approximately 0.00833. Only 0.004 rejects. In the changed scenario, ten tests were declared but only these six values are supplied. Use all ten in the denominator:

the cutoff is 0.005. Only 0.004 among the supplied values rejects; the other four decisions remain unknown.

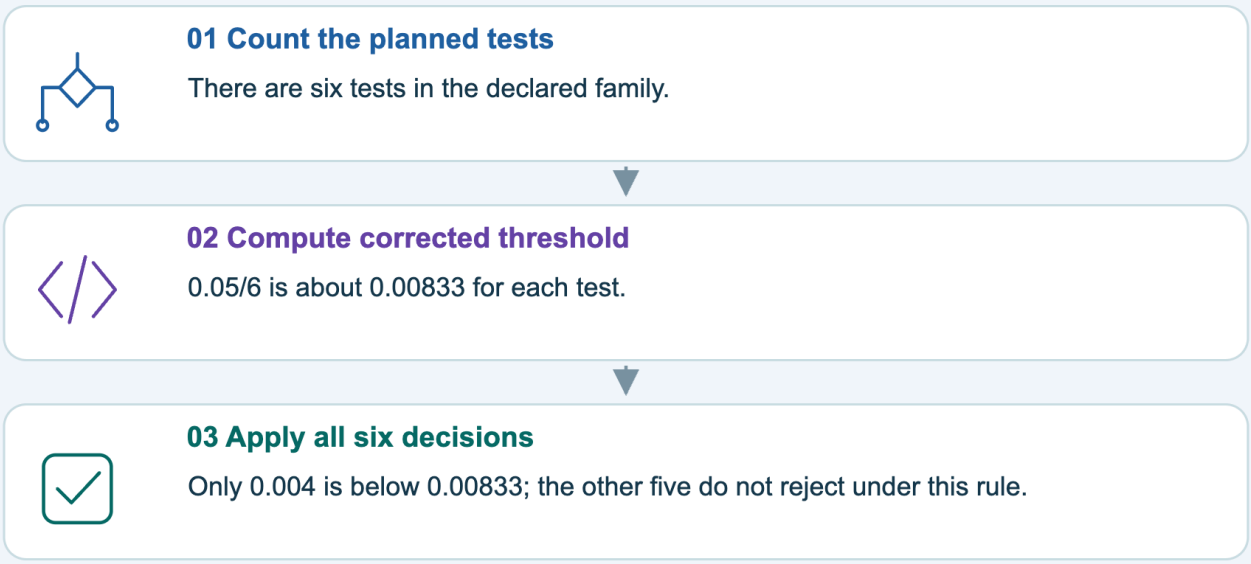
This is a comparison between declared scenarios, not permission to revise the family after seeing favorable results.

Compare the three cutoffs visually



AI ACADEMY / AI-7/15 · 02 · TRACE THE CASE

Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Count the planned tests	There are six tests in the declared family.
2. Compute corrected threshold	0.05/6 is about 0.00833 for each test.
3. Apply all six decisions	Only 0.004 is below 0.00833; the other five do not reject under this rule.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Searching many comparisons provides more chances for false positives. Bonferroni divides the error allowance across the declared family; valid individual p-values are required.

A rejected null is not automatically a large or useful improvement.

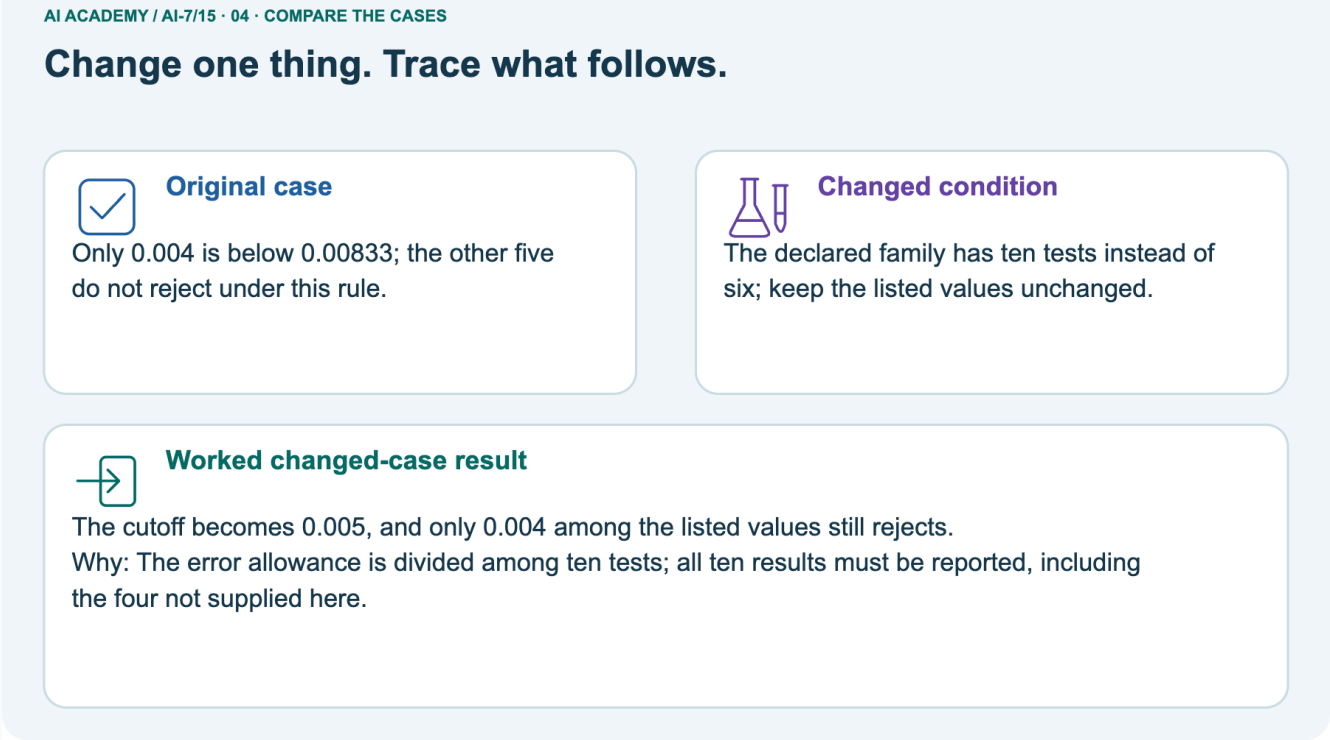
Every supplied picture decision

Separate picture test	Supplied p	Six-test decision	Ten-test decision
P1	0.004	Reject	Reject
P2	0.012	Do not reject	Do not reject
P3	0.020	Do not reject	Do not reject
P4	0.080	Do not reject	Do not reject
P5	0.400	Do not reject	Do not reject
P6	0.900	Do not reject	Do not reject
Four additional tests	Not supplied	Outside original six-test family	Unknown; remain counted in m = 10

Predict the ten-test scenario

- Change one thing: The declared family has ten tests instead of six; keep the listed values unchanged. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

State evidence limits and reporting obligations

- No effect size or real data is supplied. Bonferroni can be conservative and cannot repair invalid p-values or undisclosed outcome-driven test selection.

- Read the labelled picture: Calculate the six-test cutoff and identify which supplied p-values reject. Why must the team count every planned comparison rather than report only its smallest p-value?

▼ Earlier wording, alternatives and picture transcripts

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
def bonferroni(p_values, alpha=0.05):
    if not p_values or not 0 < alpha < 1 or any((not 0 <= p <= 1 for p in p_values)):
        raise ValueError('test settings')
    threshold = alpha / len(p_values)
    return (threshold, [p <= threshold for p in p_values])

print(bonferroni([.009, .012, .3, .4, .8]))
```

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Multiple comparisons

The facts we will use

AI ACADEMY / AI-7/15

Supplied p-values and six-test decisions

	Supplied p	At cutoff 0.00833
Test 1	0.004	Reject
Test 2	0.012	Do not reject
Test 3	0.02	Do not reject
Test 4	0.08	Do not reject
Test 5	0.4	Do not reject
Test 6	0.9	Do not reject

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

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Multiple comparisons

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Count the planned tests

There are six tests in the declared family.



2. Compute corrected threshold

0.05/6 is about 0.00833 for each test.



3. Apply all six decisions

Only 0.004 is below 0.00833; the other five do not reject under this rule.

CHANGE ONE THING

The declared family has ten tests instead of six; keep the listed values unchanged.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Repair hidden comparisons and unsupported statistical claims

> A smaller cutoff cannot rescue invalid inputs or selective reporting.

Original limitations and repair task

Adjustment cannot repair invalid tests, post-hoc outcome selection or unreported analyses.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Family, equality and validity probes

Audit question	Checked answer
p equals alpha/m exactly	Reject under the declared inclusive rule
Keep only the smallest p and call m = 1	Invalid reconstruction of the original family
Tests are dependent	Bonferroni can still control familywise error if each null p-value is valid
Individual p-values are invalid	Dividing alpha cannot repair that defect
Four family results are missing	Keep them in the family size and mark outcomes unknown
A null is rejected	Does not establish a large useful effect or the probability that the null is false
A null is not rejected	Does not prove no effect
Bonferroni is conservative	The bound can be loose; the actual error probability need not equal alpha

Write and justify a repair

- A report omits this qualification: Adjustment cannot repair invalid tests, post-hoc outcome selection or unreported analyses. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise the complete family calculation before interpreting findings

> Keep the independent four-test family for a first individual attempt.

Shared explanation

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Supported workbook trace

Use workbook W2 for the connected example "**Multiple comparisons**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual check

- Use the worked case for **Multiple comparisons**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Calculate the six-test cutoff and identify which supplied p-values reject. Why must the team count every planned comparison rather than report only its smallest p-value?

> **My explanation using a pictured step and the given facts**
>
> _____
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> _____
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> _____
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> _____

Reserve the selected four-test family

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

Run an exact family audit with explicit missing results

> Interpret every returned decision under the stated validity assumptions.

Current investigation

Five valid tests form one declared family with alpha 0.05. Their p-values are .009, .012, .3, .4, .8. Use Bonferroni: compare each p-value with 0.05/5, including equality. Show the intermediate steps and give the conclusion with its scope.

Complete executable family calculation

```
"""Exact arithmetic on stipulated valid p-values, not a fitted statistical study."""
from fractions import Fraction

def bonferroni_family(values, family_size=None, alpha='0.05'):
 if not isinstance(values,(list,tuple)) or not values:raise ValueError('supply recorded p-values')
 if family_size is None:family_size=len(values)
 if type(family_size) is not int or family_size<len(values):raise ValueError('family must include all recorded tests')
 if type(alpha) not in (str,int,Fraction):raise ValueError('use exact decimal strings or fractions')
 level=Fraction(alpha)
 if not 0<level<1:raise ValueError('alpha must be between zero and one')
 if any(type(p) not in (str,int,Fraction) for p in values):raise ValueError('use exact p-value strings or fractions')
 ps=[Fraction(p) for p in values]
 if any(not 0<=p<=1 for p in ps):raise ValueError('p-values must lie in [0,1]')
 cutoff=level/family_size
 return {'family_size':family_size,'recorded_tests':len(ps),'missing_tests':family_size-len(ps),
         'cutoff':cutoff,'reject':[p<=cutoff for p in ps],
         'adjusted_p':[min(Fraction(1),p*family_size) for p in ps]}
main=bonferroni_family(['.009','.012','.3','.4','.8'])
picture_values=['.004','.012','.02','.08','.4','.9']
picture=bonferroni_family(picture_values)
changed=bonferroni_family(picture_values,10)
boundary=bonferroni_family(['.01','.2','.3','.4','.5'])
assert main['cutoff']==Fraction(1,100) and main['reject']==[True,False,False,False,False]
assert picture['cutoff']==Fraction(1,120) and picture['reject']==
[True,False,False,False,False,False]
assert changed['cutoff']==Fraction(1,200) and changed['missing_tests']==4
```

```
assert boundary['reject'][0]
print('Main five-test decisions:',main)
print('Separate six-test picture:',picture)
print('Ten-test declared family, six supplied results:',changed)
print('Unknown four results stay unknown; no decision for them is fabricated.')
print('For valid individual tests, the union bound controls the chance of any false
rejection at most alpha, without requiring independent tests.')
print('This arithmetic does not validate supplied p-values, establish effect magnitude or
prove non-rejected nulls. No actual prompt experiment is run.')
```

Expected results and code limits

1. Predict all five main decisions before running or tracing. The full code uses exact decimal strings and rational cutoffs, checks equality, and retains the declared ten-test family when four outcomes are absent. It also calculates Bonferroni-adjusted p-values as min(1, m × p), an equivalent comparison with family alpha. None of this calculates p-values from actual observations or verifies their validity.

   The original compact floating-point example remains an optional reference; use the full exact version for boundary investigations.

Evidence record and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

```
def bonferroni(p_values, alpha=0.05):
    if not p_values or not 0 < alpha < 1 or any((not 0 <= p <= 1 for p in p_values)):
        raise ValueError('test settings')
    threshold = alpha / len(p_values)
    return (threshold, [p <= threshold for p in p_values])

print(bonferroni([.009, .012, .3, .4, .8]))
```

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: apply the rule to the new four-test family

> Calculate the new cutoff and all decisions before opening feedback.

Independent family and p-values

Four valid tests form one declared family with alpha 0.05. Their p-values are .01, .02, .2, .9. Use Bonferroni: divide family alpha by four and reject when p is at most that threshold. Compute the threshold and all four decisions.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Attempt this without hints

Four valid tests form one declared family with alpha 0.05. Their p-values are .01, .02, .2, .9. Use Bonferroni: divide family alpha by four and reject when p is at most that threshold. Compute the threshold and all four decisions.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

MAIN LESSON · SECTION 11

Save the declared family, every result and the analysis rule

> Report missing outcomes and practical-effect gaps alongside decisions.

Project and artifact

Add a labelled **multiple comparisons evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **multiple comparisons**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual local calculation or paper trace

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ Earlier wording, alternatives and picture transcripts

Evidence for your project

MAIN LESSON · SECTION 16

Explain the familywise bound and the conditions behind it

> Non-rejection and rejection both require careful interpretation.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **multiple comparisons**

> **My deciding step, evidence and one remaining question**
>
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> | |

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

Vocabulary and your explanation

Word	Meaning
Multiple comparisons	Testing several hypotheses in one declared family
Family error	The chance of at least one false rejection under relevant nulls
Bonferroni rule	Dividing a family threshold by the number of tests

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: Adjustment cannot repair invalid tests, post-hoc outcome selection or unreported analyses. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the research reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Multiple comparisons — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Five valid tests form one declared family with alpha 0.05. Their p-values are .009, .012, .3, .4, .8. Use Bonferroni: compare each p-value with 0.05/5, including equality.	For this exercise use Bonferroni: family alpha .05 divided by number of tests m; reject each test when its valid p-value is at most .05/m. This controls the family error by a bound without requiring independent tests.	Threshold .01; only .009 rejected. An unadjusted .05 rule would also reject .012 and change family control.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Six declared prompt comparisons. A team performs six valid hypothesis tests and applies a predeclared Bonferroni familywise threshold.

By the end, I can	How I will show it
Explain the mechanism and terminology of multiple comparisons	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Six declared prompt comparisons. A team performs six valid hypothesis tests and applies a predeclared Bonferroni familywise threshold.
- Which labelled step matches this detail? Only 0.004 is below 0.00833; the other five do not reject under this rule.
- 1. Apply all six decisions
- 2. Compute corrected threshold
- 3. Count the planned tests
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Apply all six decisions Only 0.004 is below 0.00833; the other five do not reject under this rule.

Searching many comparisons provides more chances for false positives. Bonferroni divides the error allowance across the declared family; valid individual p-values are required.

A rejected null is not automatically a large or useful improvement.

Changed-case question and optional hint

- Changed-case practice: The declared family has ten tests instead of six; keep the listed values unchanged. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. The cutoff becomes 0.005, and only 0.004 among the listed values still rejects.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> _____
>
> _____
>
> _____
>
> _____

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The cutoff becomes 0.005, and only 0.004 among the listed values still rejects.

The error allowance is divided among ten tests; all ten results must be reported, including the four not supplied here.

Repair and retain the record

Review Statistical tests and assumptions (ai-7/14). Goal: Explain the **mechanism** and terminology of statistical tests and assumptions. Short practice: Calculate each paired difference and the observed sum.

Count null arrangements with absolute sum at least four, compute p, and explain the decision at threshold 0.05.. Return to this week and explain what you changed.

Review Uncertainty intervals (ai-7/12). Goal: Explain the **mechanism** and terminology of uncertainty intervals. Short practice: Locate -0.8 and zero relative to the supplied interval [-1.4,-0.2].

Interpret the sign and confidence-method meaning, then describe what changes with interval [-1.4,0.3].. Return to this week and explain what you changed.

Review Experimental units and independence (ai-7/9). Goal: Explain the **mechanism** and terminology of experimental units and independence. Short practice: Count independent plants and photo views.

Construct the stated group-correct split, then explain why placing a P1 view in evaluation creates leakage for the new-plant goal.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

</details>

<details>
<summary>Reference · Six-picture concept journey</summary>

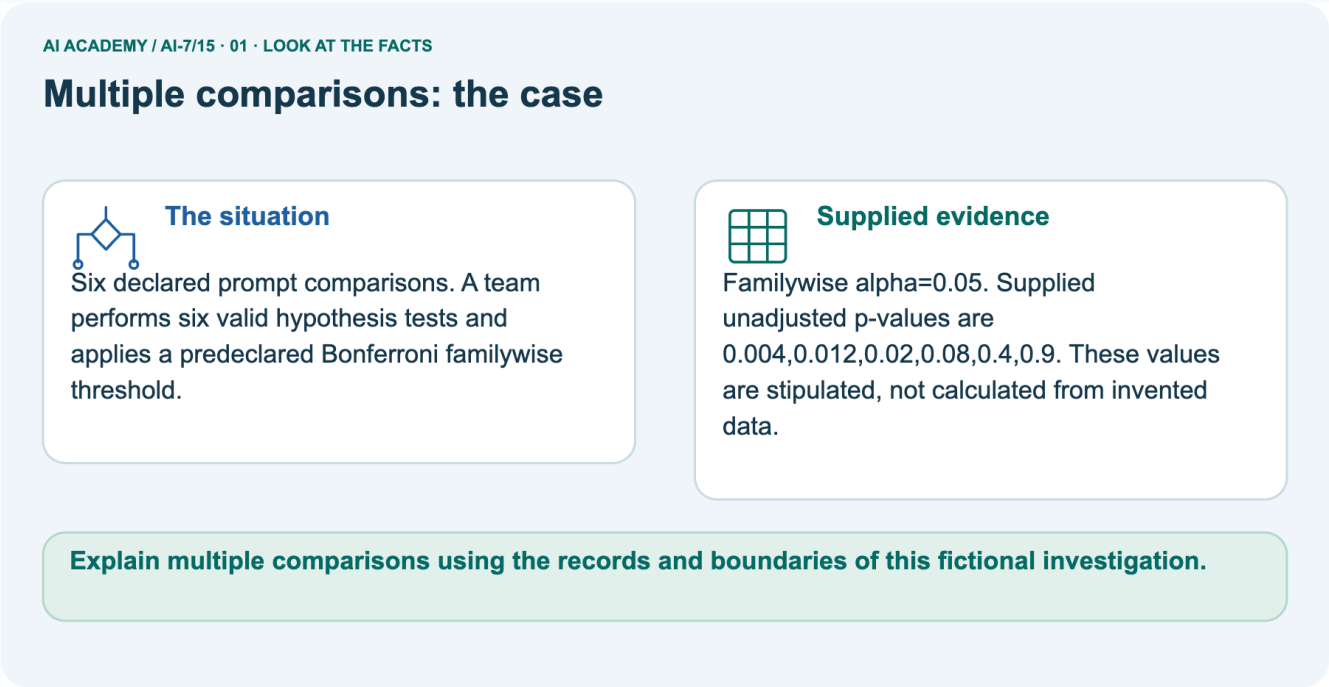
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REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Multiple comparisons: the case



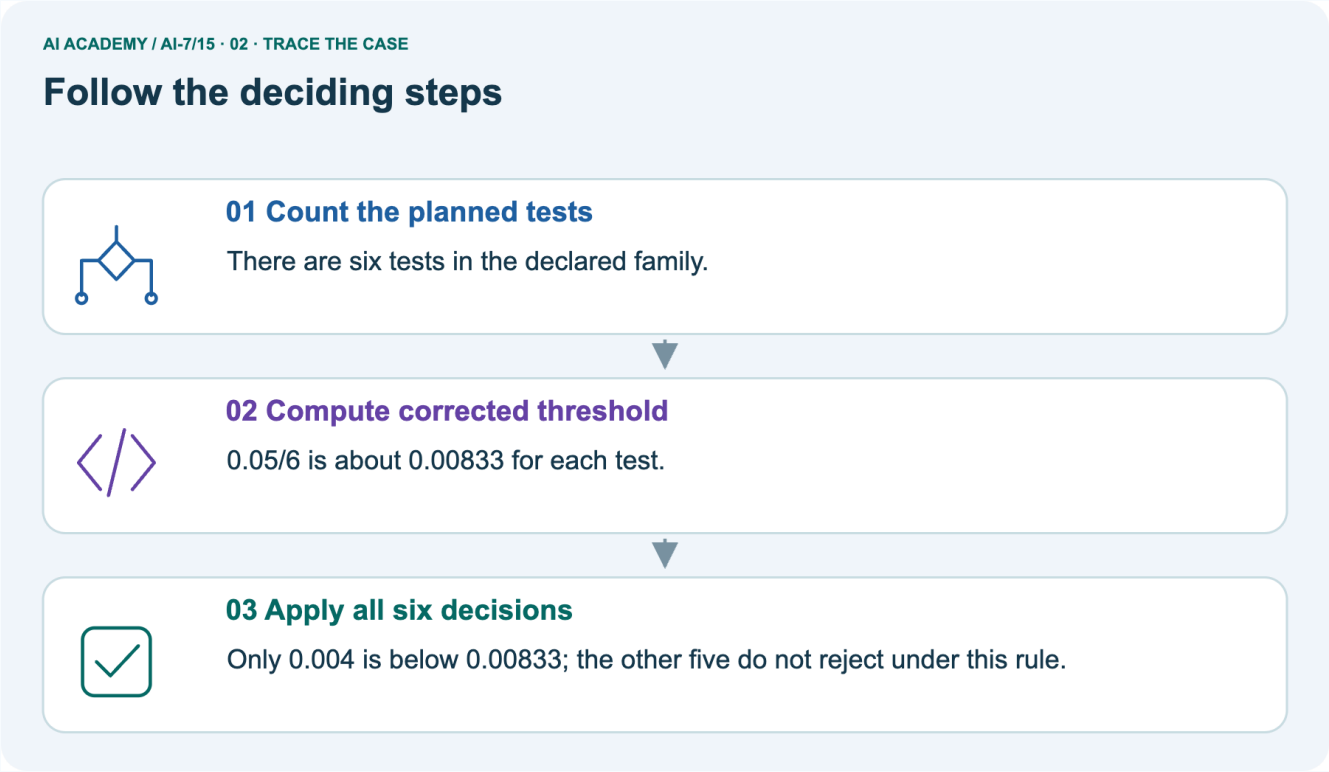
Follow the labelled picture with the matching explanation. Open the vector diagram

Six declared prompt comparisons. A team performs six valid hypothesis tests and applies a predeclared Bonferroni familywise threshold. Supplied facts: Familywise alpha=0.05.

Supplied unadjusted p-values are 0.004,0.012,0.02,0.08,0.4,0.9. These values are stipulated, not calculated from invented data.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

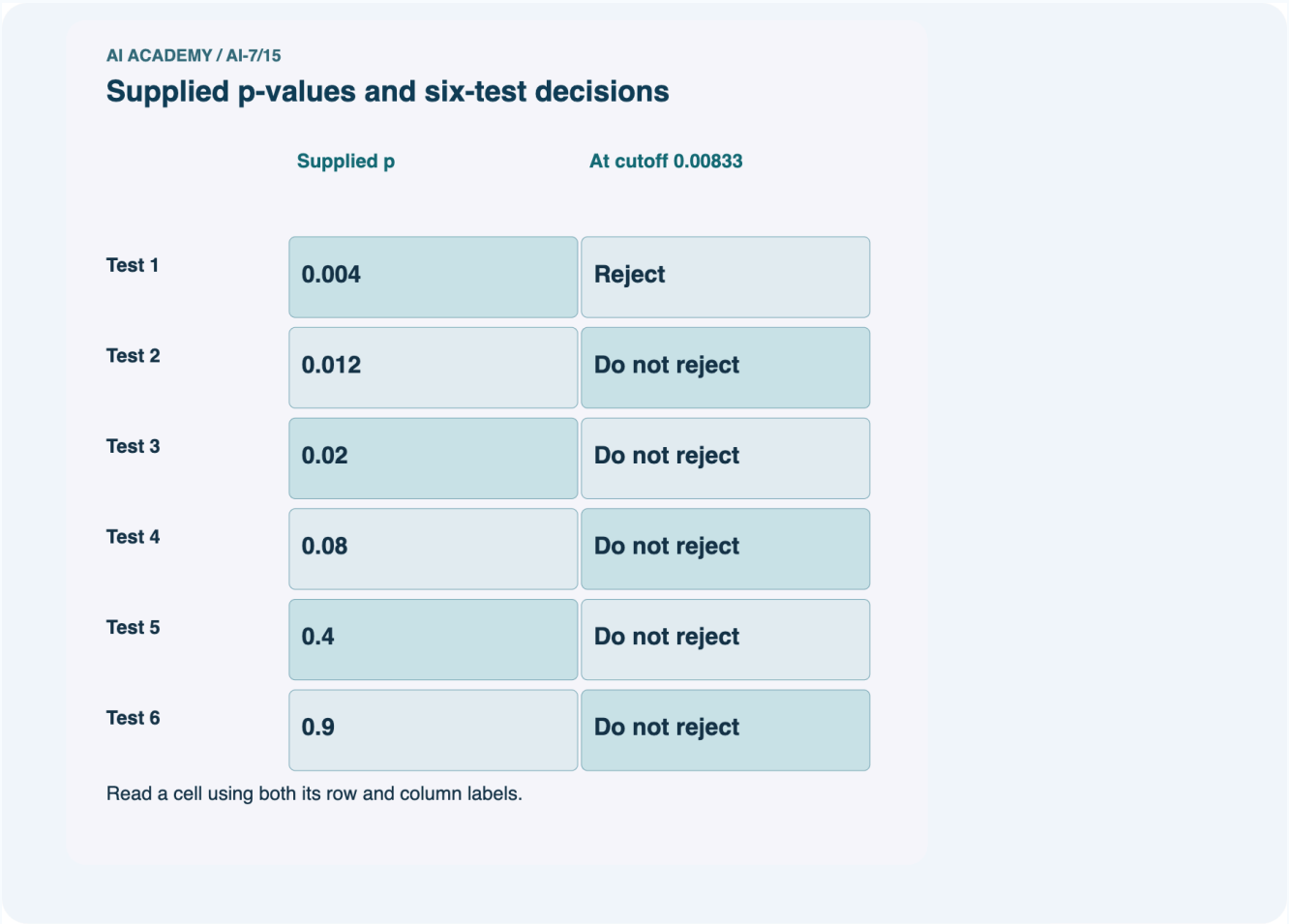


Follow the labelled picture with the matching explanation. Open the vector diagram

Count the planned tests: There are six tests in the declared family. → Compute corrected threshold: 0.05/6 is about 0.00833 for each test. → Apply all six decisions: Only 0.004 is below 0.00833; the other five do not reject under this rule.

Point to each numbered stage and explain which recorded **input** it uses.

03. Supplied p-values and six-test decisions



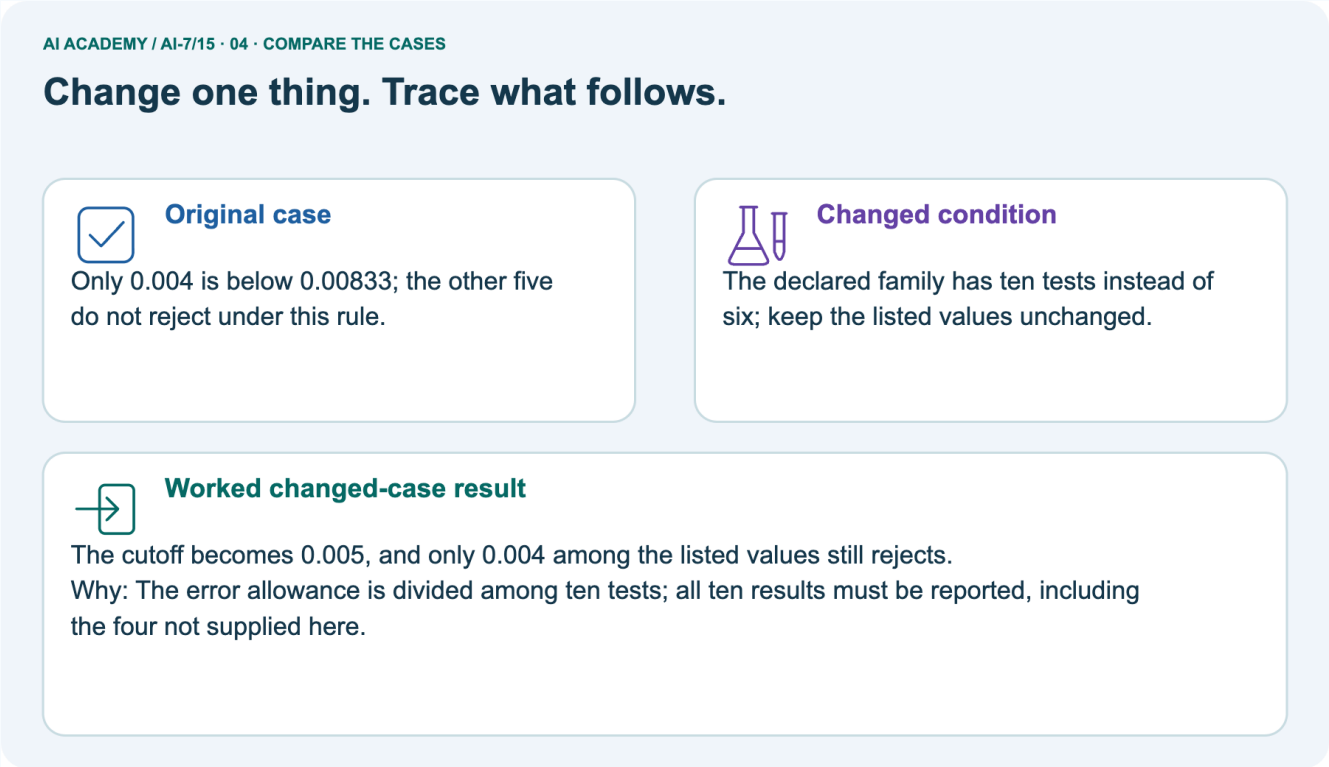
Follow the labelled picture with the matching explanation. Open the vector diagram

Supplied p-values and six-test decisions. Test 1: Supplied p = 0.004, At cutoff 0.00833 = Reject. Test 2: Supplied p = 0.012, At cutoff 0.00833 = Do not reject. Test 3: Supplied p = 0.02, At cutoff 0.00833 = Do not reject.

Test 4: Supplied p = 0.08, At cutoff 0.00833 = Do not reject. Test 5: Supplied p = 0.4, At cutoff 0.00833 = Do not reject. Test 6: Supplied p = 0.9, At cutoff 0.00833 = Do not reject.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



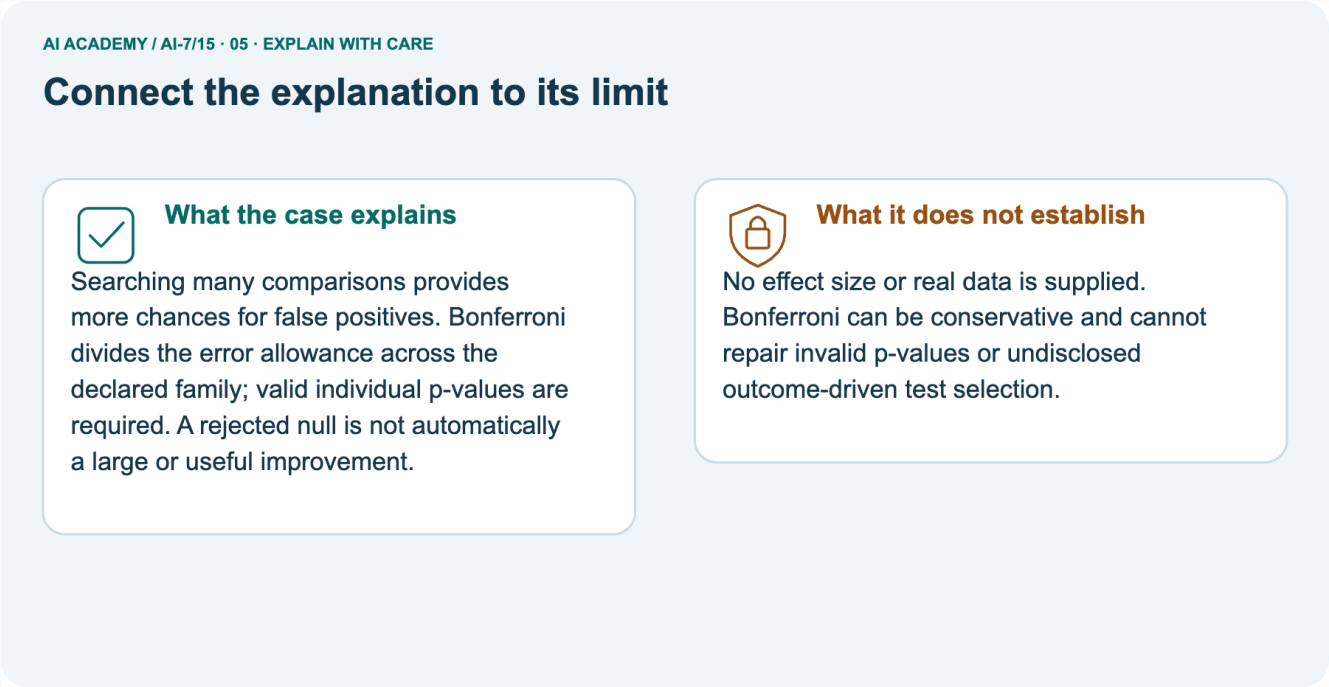
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Only 0.004 is below 0.00833; the other five do not reject under this rule. Changed condition: The declared family has ten tests instead of six; keep the listed values unchanged.

Expected result: The cutoff becomes 0.005, and only 0.004 among the listed values still rejects. Reason: The error allowance is divided among ten tests; all ten results must be reported, including the four not supplied here.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

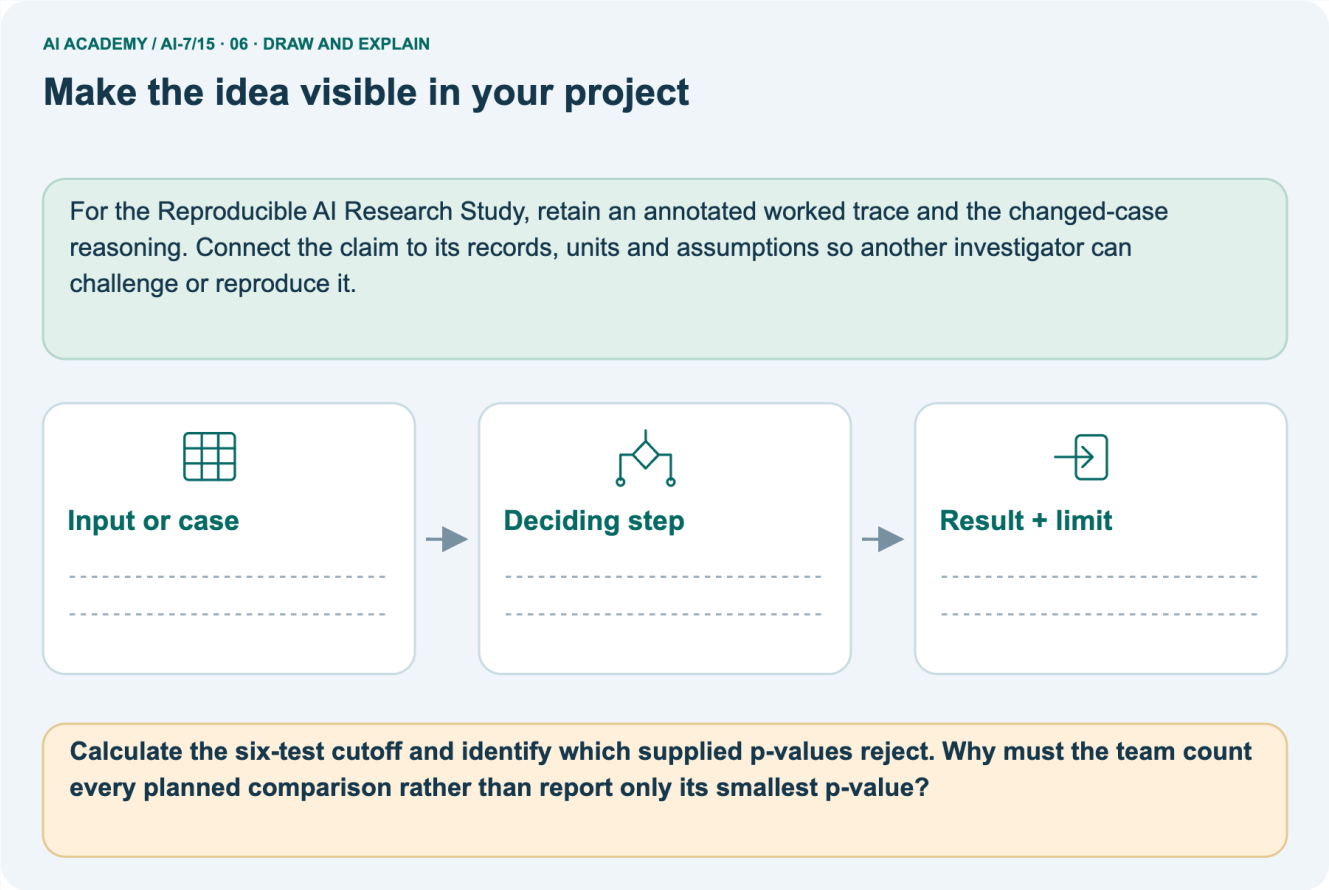
Searching many comparisons provides more chances for false positives. Bonferroni divides the error allowance across the declared family; valid individual p-values are required.

A rejected null is not automatically a large or useful improvement. Limit: No effect size or real data is supplied.

Bonferroni can be conservative and cannot repair invalid p-values or undisclosed outcome-driven test selection.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the six-test cutoff and identify which supplied p-values reject.

Why must the team count every planned comparison rather than report only its smallest p-value?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Multiple comparisons: the case

02. Follow the deciding steps

03. Supplied p-values and six-test decisions

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.